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Experimental Determination of the Charge to Mass Ratio of the Electron

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A B S T R A C T

Electrons are a recently discovered particle and little is known about their nature. While it is known that they have a charge and mass, the mass of an individual electron is not known. We attempt to find their charge to mass ratio by accelerating the electrons and then curving them with an electric field. Our results and analysis are reported.

1. Background

Leading up to the discovery of the electron by J.J Thomson in 1897 [1], many physicists were studying Crookes tubes and the mysterious cathode rays. These are glass containers with a cathode and anode at either end. When a high voltage is applied to them a ray of particles can be observed, and thus called Cathode rays as discovered by William Crookes in 1879 [4]. When a magnetic field is applied to the cathode rays it can be observed that they appear to deflect and bend. This lead J.J Thomson to the conclusion that the cathode rays are made of particles and that they are charged [2]. Using our previous knowledge of the behavior of charged particles in a magnetic field we can set the force due to the magnetic field equal to that of the centripetal force.

$$(1) \quad evB = mv^2/r$$

Building off this, charge density of the newly discovered particle can be found using a variation of the original equation in this form

$$(2) \quad e/m = v/RB$$

J.J Thomson also discovered that by applying an electric field to the particles they can be deflected in a similar manner [3]. This is especially helpful so that we can use the equation

$$(3) \quad 1/2 mv^2 = eV$$

Now we are able to find the velocity of the particle by applying a specific voltage to the system letting us change our charge density equation again to be

$$(4) \quad e/m = 2V/B^2R^2$$

Finally we can use Helmholtz coils, first proposed by Hermann Von Helmholtz [5], in order to create a uniform magnetic field that we are able to control separately from the electric field. The relevant equation for the field strength created by the coils is

$$(5) \quad B = \frac{8\mu_0NI}{\sqrt{125}a}$$

For our particular setup we know that N is 72 and a is 0.33. Finally we are capable of finding the charge to mass ratio of the particle discovered by J.J Thomson using

$$(6) \quad e/m = \left(\frac{3.91a^2}{\mu_0^2N^2}\right)\left(\frac{V}{I^2R^2}\right)$$

2. Methods

2.1 Equipment

A setup from the Welch Scientific Company was used. The center of the setup consisted of a filament that could produce electrons, a metal encasing that could accelerate the electrons, Figure(1) B, and five bars placed at varying distances from the filament, Figure(1) C. This central component was enclosed in a glass vessel under a vacuum. Surrounding the vessel were two

Helmholtz coils, Figure(1) W. The coils and the vessel were on a tiltable platform so that the Helmholtz coils could cancel out Earth's magnetic field. Three power supplies were used for powering the coils, heating the filament, and for accelerating the electrons.

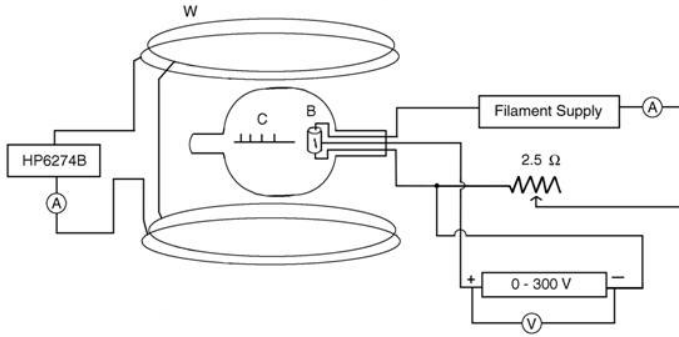


Figure 1: The experimental setup used for the measurements and calculation of the charge to mass ratio of the electron

2.2 Calculating e/m

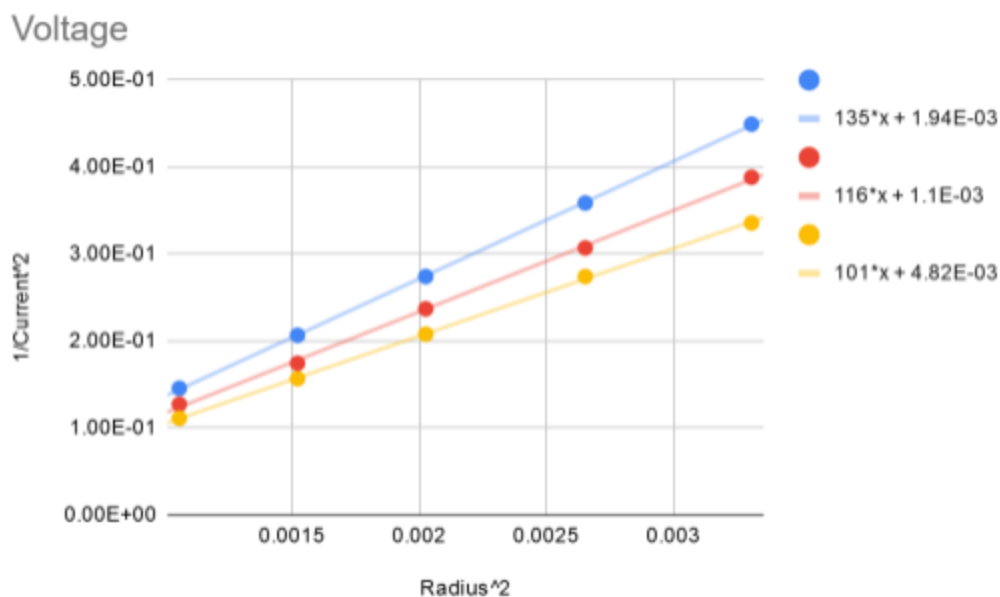
The filament was heated up and the electrons were accelerated until the electrons produced a blue glow in their path. The coils were then powered so that the electrons curved towards the bars. The current was then increased until the electrons hit the farthest bar. The current was recorded and then the process was repeated until every bar had been hit. The accelerating voltage was adjusted and all of the data was taken again multiple times. The data was then used to calculate e/m for each recorded current and its corresponding distance to peg and accelerating voltage.

3. Results

The results of the experiment are summarized in Table (1). The values from Table (1) are graphed in Graph (1). The linear fits from Graph (1) are used to calculate e/m in Table (2).

Voltage	Current	Radius
24.8	1.592	0.0575
24.8	1.77	0.0515
24.7	2.01	0.045
24.6	2.3	0.039
24.99	2.72	0.0325
29	1.705	0.0575
29.4	1.905	0.0515
29	2.154	0.045
29	2.494	0.039
28.7	2.902	0.0325
33	1.826	0.0575
33	2.01	0.0515
33	2.293	0.045
33	2.626	0.039
32.8	3.098	0.0325

Table 1. The recorded values for voltage, current, radius.



Graph 1. Linear fits for r^2 vs $1/I^2$ at each voltage

Voltage	Slope	e/m
25	135	1.76E+11
29	116	1.75E+11
33	101	1.73E+11

Table 2. Calculated values for e/m .

4. Analysis

There is some deviation from our values for e/m and the actual value of e/m . This deviation probably comes from measurement errors. One source of error would be the visual error of telling when the electrons are hitting the bar. Another source of error is Earth's magnetic field, which we may not have fully canceled out by tilting the setup.

Using our average value of e/m ($1.747E+11$) and Robert Milikan's value of the charge of an electron ($1.592E-19$), the mass of an electron can be calculated as $9.1127E-31$. This mass is minuscule compared to the mass of an atom.

5. Conclusion

The large charge to mass ratio of an electron indicates many interesting properties of electrons. For example, the Coulomb force will be many orders of

magnitude stronger than the gravitational force. Our results for e/m also indicate that charge may be quantized.

References

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